

Computer Organization and Assembly Language



Lab Manual 09

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Lab Implementation Guidelines

General Instructions

1. **Single File Structure:**
 - Implement all lab tasks within a single assembly file named task1.asm.
2. **Procedure Organization:**
 - Develop a distinct procedure for every lab task.
 - Name each procedure based on the task, for example:
 - Task1_8BitDecimalDisplay PROC
 - Task2_16BitDecimalDisplay PROC
 - Task3_DistanceConverter PROC
 - Task4_AgeConverter PROC
 - Task5_SalaryBreakdown PROC
 - Task6_CurrencyConverter PROC
 - Task7_FactorialCalculator PROC
 - Task8_MultiplicationTable PROC
 - Task9_TemperatureConverter PROC
 - Task10_PriceSplitter PROC
3. **Main Procedure:**
 - In the main procedure, invoke each task procedure sequentially.
 - After invoking each procedure, output the results using DOS interrupts or a dedicated decimal display function.
 - Sample structure:

```
text

main PROC
    call Task3_DistanceConverter
    call Task4_AgeConverter
    call Task5_SalaryBreakdown
    ...
    mov ah, 4Ch
    int 21h
main ENDP
```

4. **Input and Output Handling:**
 - Hard code each task procedure's input except `PriceSplitter`
 - Present results with clear labels (e.g., "Hours:", "Minutes:").
 - Utilize a decimal display procedure for numeric outputs when appropriate.
5. **Register and Stack Handling:**

- Save and restore any modified registers within procedures using PUSH and POP as needed.
 - Pass parameters either via registers or by pushing values onto the stack prior to the procedure call. (Don't create memory for arguments storage)
6. **Uniformity:**
- Ensure consistent formatting for input prompts and output displays across tasks.

Tasks

Task 1: Decimal Display Functions for 8-Bit

Objective:

- Output an 8-bit integer in decimal format on the screen.

Input:

- The number is provided in AL or passed through the stack.

Output:

- Display the number's decimal form using DOS interrupts.

Task 2: Decimal Display Functions for 16-Bit Numbers

Objective:

- Output a 16-bit integer in decimal format on the screen.

Input:

- The number is provided in AX or passed through the stack.

Output:

- Display the number's decimal form using DOS interrupts.

Task 3: Distance Converter (Meters to Kilometers, Meters)

Description:

- Hard code the distance in meters.
- Transform the distance into kilometers, and remaining meters.

Sample Input/Output:

- Distance: 42335 (m)
- Kilometers: 42
- Meters: 335

Task 4: Age Converter (Days to Years, Months, Days)

Description:

- Hard code the total number of days.
- Convert the days into years, months, and remaining days.

Sample Input/Output:

- Enter total days: 800
- Years: 2
- Months: 2
- Days: 10

Task 5: Salary Breakdown (Percentage Calculations)

Description:

- Divide a total salary into various components based on percentages.

Instructions:

1. Hard code the total salary.
2. Compute:
 - Basic = 40% of total
 - House Rent = 30% of total
 - Conveyance = 20% of total
 - Other Allowances = 10% of total

Sample Input/Output:

- Enter total salary: 10000
- Basic: 4000
- House Rent: 3000
- Conveyance: 2000
- Other Allowances: 1000

Task 6: Currency Converter (PKR to USD, EUR, GBP)

Description:

- Convert an amount in Pakistani Rupees (PKR) to USD, EUR, and GBP using predefined exchange rates.

Instructions:

1. Hard code the amount in PKR.
2. Apply integer division for conversions:
 - $\text{USD} = \text{PKR} / 280$
 - $\text{EUR} = \text{PKR} / 300$
 - $\text{GBP} = \text{PKR} / 350$
3. Output the converted amounts.

Sample Input/Output:

- Enter amount in PKR: 5600
- USD: 20
- EUR: 18
- GBP: 16

Task 7: Factorial Calculator (For Numbers 1–8)

Description:

- Compute the factorial of a number between 1 and 8.

Instructions:

1. Hard code a number n (where $1 \leq n \leq 8$).
2. Display the computed factorial.

Sample Input/Output:

- Enter a number (1-8): 5
- Factorial: 120

Task 8: Multiplication Table Generator

Description:

- Produce a multiplication table for a specified number (1–10).

Instructions:

1. Hard code a number.
2. Output the table.

Sample Input/Output:

- Enter number: 7
- $7 \times 1 = 7$
- $7 \times 2 = 14$
- ...
- $7 \times 10 = 70$

Task 9: Temperature Converter (Celsius to Fahrenheit and Kelvin)

Description:

- Convert a temperature from Celsius to Fahrenheit and Kelvin.

Instructions:

- Input: Hard code temperature value in Celsius.
- Compute Fahrenheit: $F = (C * 9 / 5) + 32$
- Compute Kelvin: $K = C + 273$
- Output both results using integer operations.

Task 10: Price Splitter

Objective:

- Develop an assembly program to assist a group in dividing a total cost using various methods. Incorporate procedures, DIV, MUL, and conditional statements to determine individual shares.

Program Specifications:

1. **User Input:**
 - Total cost (as an integer).
 - Number of persons (between 1 and 5).
2. **Menu for Cost Division Methods:**
 - **Equally:** Split the total cost uniformly among all persons.
 - **By Share:** User provides individual shares for each person.
 - Ensure the sum of shares matches the total cost.
 - **By Percentage:** User specifies percentages for each person.
 - Confirm that percentages sum to 100%.
3. **Procedures:**
 - Assign a separate procedure to each division method.
 - The main program invokes the selected procedure based on the user's choice.

4. **Output Display:**

- Present each person's calculated cost in an organized format.
- Example:
 - Total Cost: 500
 - Number of Persons: 3
 - Choose division method:
 1. Equally
 2. By Share
 3. By Percentage
 - Option: 1
 - Person 1: 167
 - Person 2: 167
 - Person 3: 166